=== NOAA AWS Parallel Cluster ===

Run JEDI (FV3-JEDI)

Note) If you have already set up the repository and input dataset on Day 1, please skip the first four steps (1 - 4).

1. Clone the following repository:

```
git clone https://github.com/chan-hoo/cadre26_noaa_tutorial
```

2. Build the GDAS App:

```
cd cadre26_noaa_tutorial

./build_gdas_cluster.sh
```

Note) The building process takes about 20 minutes on the cluster.

3. Check if the JEDI executables ('fv3jedi_var.x' and 'gdas_fv3jedi_fv3inc.x') exist in the following path:

```
cd ../GDASApp

# to check the build status:

tail -n 10 build.log

# check executables in bin dir:

cd build/bin

ls

cd ../../../cadre26_noaa_tutorial
```

4. Download the JEDI input dataset from S3:

```
wget https://s3.us-east-1.amazonaws.com/epic.sandbox.content/cadre26-input-data.tar.gz

tar -xvzf cadre26-input-data.tar.gz

# You can see the data directory "cadre26". Remember the path "/path/to/cadre26/input_data"

# You should update JEDI_INPUT_PATH="/scratch/cadre26/input_data" in Step 6 with it.

cd cadre26/input_data

pwd

cd ../../
```

5. Copy the JEDI input yaml files to the specific location:

```
cd input_yaml

cp Day1/jedi_3dvar* .

(or)

cp Day2/[exp_case]/jedi_3dvar* .

(or)

cp Day3/[exp_case]/jedi_3dvar* .
```

6. Open the job submission script and modify it as needed:

```
cd ..

vim run_3dvar_cluster.sh

(Check your account, JEDI_BIN_PATH, JEDI_INPUT_PATH, EXP_NAME_BASE)

:wq
```

7. Submit your job:

```
sbatch run_3dvar_cluster.sh
```

8. Check your job:

```
squeue -u ubuntu (or your username)
```

Note) Remember the job ID of your current job.

9. Move to the experimental case directory for your job:

```
cd exp_case/[EXP_NAME_BASE].[job_ID]
```

10. Check the log files:

```
vim OUTPUT.fv3jedi

Vim errfile_fv3jedi
```

11. Move to the output directory:

```
cd output
ls
```

```
(plot_pyenv) ubuntu@ip-10-29-93-229:~/cadre26_noaa_tutorial/exp_case/cadre26.132/output$ ls
cubed_sphere_grid_atminc.jedi.nc          obs.20240224.t00z.atms_n20.satbias_cov.nc
diag.atms_n20_2024022400.nc               ufsda.t00z.atminc.cubed_sphere_grid.tile1.nc
diag.conventional_ps_2024022400.nc        ufsda.t00z.atminc.cubed_sphere_grid.tile2.nc
diag.gnssro_cosmic2_2024022400.nc         ufsda.t00z.atminc.cubed_sphere_grid.tile3.nc
diag.satwnd.abi_goes-16_2024022400.nc     ufsda.t00z.atminc.cubed_sphere_grid.tile4.nc
diag.scatwnd.ascat_metop-b_2024022400.nc  ufsda.t00z.atminc.cubed_sphere_grid.tile5.nc
obs.20240224.t00z.atms_n20.satbias.nc     ufsda.t00z.atminc.cubed_sphere_grid.tile6.nc
```

Run Diagnostics

Note) If you have already set up the conda environment and you have cloned the ufs_da_diagnostics repository, please skip the following two steps (1 and 2).

1. Clone the diagnostic tools from GitHub:

```
cd cadre26_noaa_tutorial

git clone https://github.com/jkbk2004/ufs_da_diagnostics
```

2. Set up the conda environment:

```
# The following pip command only needs to run once

cd ufs_da_diagnostics

pip install -e .

pip install seaborn scipy
```

3. Update the input YAML files for the diagnostic tools:

```
cd diagnostic/yamls/[exp_case]

vim increment_maps.yaml

# change “prefix”s under “experiments”

:wq

vim obs_diag.yaml
```

```
# change “diag”s to the paths of your output

:wq

vim spectra_bkg_inc.yaml

# change “atm_file” under “background” and “prefix” under “increments”

:wq

vim spectra_ana_inc.yaml

# change “prefix”s under “experiments”

:wq
```

4. Create a test directory:

```
cd ../../..

mkdir -p test

cd test
```

5. Run the diagnostic tools one by one:

Day1:

```
ufsda-inc-maps  --yaml ../diagnostic/yamls/day1/increment_maps.yaml

ufsda-obs-diag  --yaml ../diagnostic/yamls/day1/obs_diag.yaml

ufsda-spectra-bkg-inc  --yaml ../diagnostic/yamls/day1/spectra_bkg_inc.yaml

ufsda-jedi-log ../exp_case/[EXP_NAME_BASE].[job_ID]/OUTPUT.fv3jedi --output

day1_log_report.txt
```

Day2 (nicas):

```
ufsda-inc-maps  --yaml ../diagnostic/yamls/day2_nicas_length_scale/increment_maps.yaml

ufsda-obs-diag  --yaml ../diagnostic/yamls/day2_nicas_length_scale/obs_diag.yaml

ufsda-spectra-ana-inc  --yaml ../diagnostic/yamls/day2_nicas_length_scale/spectra_ana_inc.yaml

ufsda-jedi-log ../exp_case/[EXP_NAME_BASE].[job_ID]/OUTPUT.fv3jedi --output
day2_log_report.txt
```

Day3 (error):

```
ufsda-inc-maps  --yaml ../diagnostic/yamls/day3_atms_err/increment_maps.yaml

ufsda-obs-diag  --yaml ../diagnostic/yamls/day3_atms_err/obs_diag.yaml

ufsda-spectra-ana-inc  --yaml ../diagnostic/yamls/day3_atms_err/spectra_ana_inc.yaml

ufsda-jedi-log ../exp_case/[EXP_NAME_BASE].[job_ID]/OUTPUT.fv3jedi --output
day3_log_report.txt
```

Download File to Your Laptop

1. Set up two private keys:

```
# Create the private key file “epic_workshop.pem” (w/ Kris)

ssh-add ~/.ssh/epic_workshop.pem

ssh-add ~/.ssh/[your other private key file name]
```

Note) If you have already set up the private keys, please skip this step 1.

2. Create a compressed file:

```
cd ../  (parent directory of the “test” directory)

tar -cvzf test.tar.gz test

# “test.tar.gz” file will be created in the current path.
```

3. Download the compressed file:

```
scp -J [user_name]@jump.epic.noaa.gov ubuntu@cadre:[path to the file] .

tar -xvzf test.tar.gz
```

Note) [user_name] may be different from your cluster user_name.